**Title: An intelligent service-based system composing web services**

**Supervisor :** Amel Benna, DSISM, CERIST

**Summary:**

The evolution of distributed computing, IoT, and AI has introduced new challenges in service composition beyond traditional concerns such as discovery, interoperability, and QoS. Modern compositions must be highly dynamic and context-aware, adapting autonomously to changing environments, user needs, and service availability. The growing scale and heterogeneity of services demand intelligent, semantics-driven automation, while ensuring end-to-end security, privacy, trust, and regulatory compliance across multi-provider ecosystems. Additionally, compositions must scale efficiently across cloud-edge infrastructures, remain explainable and maintainable, and balance complex non-functional requirements such as performance, cost, and energy efficiency. Addressing these challenges is crucial to building robust, adaptive, and intelligent service compositions in next-generation distributed systems.

The goal of this internship is to implement a prototype of system automating the description and composition of services that must be context-aware, adapting autonomously to changing environments, user needs, and service availability. Services' should continuously adapt to changing user needs, network conditions, environmental contexts (e.g., location aware services), and application requirements. The description of services is enhanced with annotations that are captured from the service's previous interactions with other services. The structure of these annotations is based on a specific model adopted from [1], ensuring consistency between the different descriptions. Additionally, this model is enhanced with contextual and privacy information. 3 types of annotations are targeted: annotation related to interaction or not with other services, interactions context. and policy associated to the use of these annotations.

**Key Words:** annotations, Web services, Composition, LLM techniques.

This project aims at implementing a system that:

1- Automate the process of annotating the services so that a service supports self-configuration, self-adaptation, and self-protection capabilities.
2- Automate the process of service composition based on services' annotations with 2 solutions:
    a. Implement a service-based system by composing existing web services to provide integrated functionality
    b. Implement an intelligent service-based system by composing conventional web services with LLM-based services to deliver adaptive and context-aware functionalities. This system that integrates LLM capabilities should combine:
        i. Traditional service composition (integration, orchestration, APIs)
        ii. LLM components (data-driven or intelligent services for reasoning, automation, or dialogue)
    c. Compare and discuss between the results of the 2 solutions

**References:**

[1] A. Benna, Z. Maamar, M. Ahmed-Nacer: A MOF-based Social Web Services Description Metamodel. MODELSWARD 2016: 217-224